NEC 2026 - E-Cell IIT Bombay

Participant Submission

Name: Riya Deshmukh
Team: Titan - Energy & Nanomaterials

My Motivation

In our materials research lab, we investigated dust accumulation on solar panel arrays in dry regions like Rajasthan. Dust buildup reduces energy output by up to 20% in just two weeks, requiring massive volumes of water for manual washing--an ironic waste in water-scarce arid zones.

We formulated a hydrophobic, self-cleaning nano-coating that prevents dust adhesion without degrading optical clarity. My motivation is to move this coating out of liquid vials in our lab and onto utility-scale solar farms. Solving clean energy efficiency challenges requires practical chemical engineering applied to existing infrastructure.

My Expectations from E-Cell

B2B industrial sales cycles for solar farm operators are notoriously long and require extensive field testing data. I expect E-Cell to help us structure corporate pilot agreements and handle intellectual property licensing.

I want NEC to give us exposure to industrial energy mentors who can critique our cost per square meter, coating longevity metrics, and execution strategy. I also hope to connect with business-minded co-founders who complement our technical research background.